



KENYATTA UNIVERSITY

DEPARTMENT OF ENGLISH

AEN 210:INTRODUCTION TO PHONETICS AND PHONOLOGY OF
ENGLISH

1.1 Introduction

1.2.1 What is phonetics?

Phonetics is a discipline that deals with the systematic study of all speech sounds which human beings are capable of producing. It provides means of identifying, recognizing, describing and classifying virtually all the speech sounds that can be produced by the human vocal tract

1.2.2 Phonetics and Phonology; Phonetics differs from phonology. **Phonology**

is a branch of linguistics that investigates the ways in which speech sounds are used **systematically in different languages to form words and utterances**. Thus, phonology considers speech sounds as they *occur in particular languages* whereas phonetics generally deals with all human speech sounds. However, one must have knowledge of basic concepts of phonetics in order to understand phonology. Put differently, phonetics provides tools essential in undertaking any study in phonology. In this way, the two studies are intertwined and most of the time it is not easy to clearly distinguish one discipline from the other. For instance, when dealing with the issue of vowels, an analyst may not know whether the issue clearly falls in the domain of Phonetics or Phonology. This is because in order to

describe various English vowels for instance, one will use the three parameters i.e. tongue height, the horizontal axis of the tongue, and shape of the lips.

1.2.3 Branches of phonetics

There are three main branches of phonetics:

Acoustic Phonetics

This is a branch of phonetics that deals with the physical properties of speech sounds. Laboratory instruments are often used to measure various properties of sound segments. As an example, electric equipment such as sound spectrographs are used to register the time taken to articulate a speech sound, its pitch, its intensity, etc.

Auditory Phonetics

This branch of phonetics examines how sound waves are received and interpreted by the human ear in the process of communication. **Articulatory Phonetics**

This is the third branch of phonetics. It deals with the processes involved in the production of sound segments. It is this branch that we shall focus on in this unit because most of the phonological theories that we shall be studying in other linguistic units in future are based on articulatory phonetics, and not on the first two branches.

1.2.4 Uses of phonetics

Phonetics has many applications in life.

- a) Teachers of language benefit from a study of phonetics. For example, when testing the reading fluency of learners, they need to diagnose the pronunciation errors made, and find ways of correcting them. This is particularly crucial in many second-language learning situations. In the Kenyan context, many teachers will find it difficult to teach the English language without both practical and theoretical knowledge of phonetics.
- b) Linguists also need a knowledge of phonetics. They can only carry out



serious research if they are well grounded in phonetics. In other levels of linguistics, such as Phonology, Morphology and Syntax, questions of phonetics often arise. For example, in the study of the English Plural morpheme, one must give a phonetic explanation.

C) Speech therapists also need phonetics. They need to understand how various vocal tract organs work in the production of speech sounds. It is only after acquiring phonetic knowledge that they can diagnose and treat speech disorders.

d) Phonetics is indispensable to communication and computer engineers, actors, philologists, radio/TV broadcasters, and others. Communication and computer experts need some training in phonetics in order for them to perform their jobs effectively. Actors, particularly comedians, have to identify and imitate certain accents so as to entertain their audiences; thus, they need practical phonetic skills. Philologists may not be able to study the laws of sound changes among related languages.

The above clarifies that a study of phonetics prepares you for more than just teaching language. Opportunities abound with this knowledge. All one needs to do is to be creative and utilize his knowledge!

2.2.1 What is Articulatory Phonetics?

There are a number of linguists who have defined articulatory phonetics and the different aspects, methods and approaches that are used in the study of articulatory phonetics. We will look at four of these definitions.

Fromkin, Rodman and Hyams (2009) define **articulatory phonetics** as *the branch of phonetics that studies how the human vocal tract produces the sounds of language*.

Yule (2010) defines articulatory phonetics, as the branch of phonetics, which studies how speech sounds are produced, or articulated.

Hayes (2009) defines articulatory phonetics as *the branch of linguistics that deals with how sounds are produced in the human vocal tract*.

Crystal (2008) defines articulatory phonetics as “the study of the way speech sounds are made (‘articulated’) by the vocal organs.”

2.2.2 The Vocal Tract

The organs, which take part in the production of speech sounds, are called **speech organs** (Clark, Yallop and Fletcher, 2007). When we are speaking, a stream of air comes out from the lungs and is obstructed at various places for the production of sounds. Sounds cannot occur without air. The term "**vocal tract**" refers to all the portions of the human anatomy through which air flows in the course of speech production (Hayes, 2009). The parts of the body have other primary biological functions. Speech is a secondary function that has been superimposed on them. These include: *the lungs*, *the larynx* (commonly referred to as "voice box"), *the glottis* (vocal cords), *the pharynx*, *the tongue*, *teeth*, *lips*, *palate* (hard roof of the mouth), *velum* (soft roof of the mouth) and *the nose*.

The organs of speech fall into three groupings:

- i. Respiratory system: Lungs generating air stream
- ii. Phonatory system: larynx and vocal folds
- iii.

Articulatory system: the vocal tract

Source:

Source: Hayes (2009:2) Parts of the Tongue

The tongue is somewhat deceptive in its size and shape. The parts that are obvious to an external observer are the tip (sometimes called the apex) and the blade. These are merely an appendage to the much larger tongue body (also called dorsum), a roundish muscular body that can move in all directions. Movements of the dorsum can radically change the shape of the vocal tract, a fact that is crucial in the production of distinct vowel sounds. The rear surface of the dorsum is called the tongue root.

4.2.1 What is Phonation?

Phonation refers to the status of vocal folds while air from the initiatory airstream is passing through the glottis. It is the rapid opening and closing of the larynx for sound producing airstream to pass. According to Catford (1988, p. 218), phonation is an activity in the larynx which is neither initiatory nor articulatory in which the air stream is modulated by its passage through the glottis (the space between the vocal cords) before being finally 'shaped' into a specific sound by the articulation. This process leads to the production of two types of consonants: **voiced** and **voiceless consonants**.



Voiceless consonants lack of any phonation due to the state of the vocal folds that are apart. The resulting sounds are voiceless. Air passes unobstructed through the glottis and the following are some of the sounds that are produced: [p], [t], [k], [f], [s] in English. The English minimal pair *wrangle* voiced /g/

vs. *wrankle* voiceless /k/ show the voice vs. voiceless contrast. In Kiswahili the minimal pair that illustrates the voice vs. voiceless contrast is the pair *piga* /g/ voiced vs. *pika*/k/ voiceless.

The figure below shows the state of the vocal fold during voicing.

Source: O’Grady, Dobrovlsky and Katamba (1997)

Ladefoged (1971), Laver (1980), and Ni Chasaide & Gobl (1995) point out that breathy phonation is characterized by vocal cords that are fairly abducted (relative to modal and creaky voice) and have little longitudinal tension. There are languages that distinguish phonemically breathy voiced and regular modal voiced sounds. Some of these languages, e.g., Hindi, Newar, Tsonga, make this contrast among their nasals. Words illustrating the breathy vs. modal voiced contrast in Newa.

Vocal folds are held loosely open, but flap as air rushes by. Usually heard on release of following vowel Used “dotted” diacritic **Voiced sounds include:**

1. Creaky (creaky voice, laryngealization)

According to Titze (1994; 2008), creaky voice which is also known as laryngealization, pulse phonation, vocal fry, or glottal fry occurs when the [arytenoid cartilages](#) in the [larynx](#) are drawn together. The [vocal folds](#) are compressed rather tightly, becoming relatively slack and compact. They vibrate irregularly at 20–50 pulses per second, and the airflow through the [glottis](#) is very slow. Although a creaky voice may occur with very low pitch, as at the end of a long [intonation unit](#), it can occur with any pitch. Creaky sounds are characterized by low frequency vibration of the vocal cords as one end of the vocal folds is closed completely, while a small opening in the front is opened; it vibrates slowly and produces a low frequency sound. The vibration occurs in the anterior part and the flow rate of 12-20 cc/s

According to, (Ladefoged 1971, Laver 1980, Ní Chasaide and Gobl 1995), creaky voice, contrasts with modal voice in many languages, with both modal voice and breathy voice in other languages. Creaky phonation (also termed vocal fry) is typically associated with vocal folds that are tightly adducted but open enough along a portion of their length to allow for voicing.

2. Whisper

There is a greater constriction of the glottis (>25% of the possible glottal area) which results from the abduction of the vocal folds, an opening between the arytenoid cartilages allows the air to escape. The characteristic flow rates about 25-30 cc/s. The linguistic uses of whisper mode are quite rare; however, the paralinguistic use may be of signals, secrecy or confidentiality. Only voiced sounds can be whispered. **5.2.1 Place of articulation**

Consonants are produced by obstructing, in some way, the flow of air through the vocal tract. We can identify the place where the obstruction takes place, and the vocal tract organ (speech organ or articulator) involved.

Articulators are either active or passive. The active speech organ moves towards a passive one (that does not move).

1. Bilabial

This describes sounds made with two lips. The lower lip moves towards the upper one and together they form a stricture of complete closure.

Examples are [p], [b] and [m]. Other examples include

2. Labio-dental

These are sounds produced when the lower lip and the upper front teeth come into contact. Sounds include: [f], [v] and [ɸ] as in amphetamines

3. Inter-dental

These sounds are produced when the tongue forms a stricture with the upper front teeth. Examples: [θ] in 'think' and [ð] in 'these'

4. Alveolar

In the production of these sounds, the tongue moves to the alveolar ridge and together they form a stricture. Examples:

[t], [d], [n], [s], [z], [l], [r], [ʀ]



5. Retroflex

The place is the same as alveolar. However, the tip of the tongue curls back, past the alveolar ridge. Examples: [ɳ], [ʂ], [ʈ], [ʣ], [ɻ], [ɭ], [ɽ], [ɳ], [ʂ], [ʈ], [ʣ], [ɻ], [ɭ]

6. Palato-alveolar

These are sounds articulated when the tongue moves towards the alveolar ridge region and the front of the hard palate. Examples: [tʃ], [dʒ], [tʃ], [dʒ] 3
[ʃ] 3

7. Palatal

The tongue moves to the hard palate and forms a stricture of some sort. Examples: [j], [ç], [ç], [ç]

8. Velar

The sounds are produced when the tongue touches the soft palate (velum). Examples: [k], [g], [ŋ], [X], [ɣ], [ɰ], [L].

9. Uvular

The sounds are produced when the tongue is retracted so that it touches the uvula. English does not have such sounds. Examples: [x] khusoma-Luhyia; [ʁ], [ħ]

10. Pharyngeal

In the production of these sounds, the walls of the pharynx act as the articulators. English does not have such sounds. Examples for sounds in Arabic: [ʕ]

11. Glottal

The articulators are the vocal cords which are found in the larynx. Examples [h], [ʔ].

5.2.2 Manner of Articulation

The organs involved in the production of a sound segment are called articulators. The term ‘**manner of articulation**’ refers to the various ways in which the articulators form a channel through which the air stream is interfered with in producing a consonant sound.

In the production of a sound, the articulators may:

- Completely shut off the vocal tract causing a complete blockage



- Partially shut off the vocal tract resulting in a 'narrow' passage

The different manners of articulation are: **stop, fricatives, affricates, laterals, trills and taps, and approximants.**

1 Stops.

This manner of articulation can be found in the production of such sounds as [p], [t], [k], [m], [n] in the initial sound segments in the words 'pill, till, kill, mill, and nil'.

There is complete closure of the vocal tract as you try to articulate those sounds. Air is momentarily, completely blocked from escaping. Consequently, pressure builds up behind the closure as the air in the vocal tract continues flowing. When the articulators separate abruptly, the pressure that had built up suddenly bursts out with a small 'explosion'. The sounds produced in this manner are called STOPS.

Stops are consonant sounds produced with a stricture of complete closure.

There are two types of stops:

a) **Oral stops** (also called **Plosives**)

In their production, the velum is raised closing the entrance to the nose (velaric closure). Air flows solely through the mouth. Two articulators completely shut off the air stream in the vocal tract. When the articulators separate abruptly, the air is released with an 'explosion' through the oral cavity. b) **Nasal stops**

In the production of these sounds, the velum is lowered. The air stream is completely blocked in the oral cavity. When the articulators separate suddenly, the air is shunted through the nasal cavity.

2 Fricatives

In the articulation of these sounds, the articulators are brought very close together leaving a very narrow passage through which air squeezes on its way out, producing some audible friction sound. Since the air stream is partially obstructed and causes turbulence as it tries to escape, the sounds produced are called **FRICATIVES**.

All fricatives are oral and are produced with a stricture of close approximation.

The sounds include [f], [v], [θ], [ð], [s], [z], [ʃ], [ʒ] 3 Others include:

The bilabial fricatives [β] and [β̥]



Retroflex fricatives [ʃ], [ʒ] ʒ

Velar fricatives [x], [χ]

Palatal fricatives [ç], [j]

Uvula fricatives [ʁ], [ħ]

Pharyngeal fricatives [ʕ]

Glottal fricatives [h], [ʕ]

3 Affricates

In the production of affricates, first the articulators come together and completely cut off the flow of the air from the vocal tract (like for stops), then they gradually separate and the air is released through a narrow passage causing some audible friction (like for fricatives).

Thus, an affricate is a sequence of stop followed by affricative:

stop + fricative = affricate

Examples of affricates include [tʃ] in easch and [dʒ] in judge

4 Laterals

To produce a lateral, there must be an incomplete closure between one or both sides of the two articulators. Air is obstructed at a point along the Center of the mouth but the sides of the tongue are left low and air is allowed to escape over both sides of the tongue. The type of stricture is referred to as an intermittent stricture. In the production of [l] in lead, the tongue feels colder on the sides of the tongue that are not in contact with the roof of the mouth, an indication that air is escaping on the sides.

5 Trills and Taps

Trills are produced when one articulator is held loosely near the other and the flow of air between the two articulators causes repeated vibration. Examples of trills are [r] alveolar trill, [ʙ] bilabial trill, [ʀ] uvular trill.

Taps (also known as Flaps) are caused by a single minute striking movement of one articulator over another. Examples are [ɾ] alveolar tap, [ɰ] retroflex tap

Trills and taps involve neither a stricture of complete closure as in stops nor that of close approximation as in fricatives. Furthermore, the intermittent stricture type used in the production of laterals is not employed in the articulation of trills and taps. Instead, an alternation of between all three stricture types is maintained.

6 Approximants

In this articulation, the articulators are close to each other but not close enough to produce turbulence. The degree of ‘narrowness’ between the two articulators is reasonable so that air flows rather freely out of the vocal tract.

Example, [j] in the word ‘you’. The tongue and the hard palate are close to each other but they form an open stricture which makes it possible for air to escape without causing an audible friction sound. Other approximants are:

- [w] Velar approximant
- [ɥ] Labio-dental approximant
- [ɻ] Retroflex approximant
- [ɰ] Velar approximant.

5.2.3 The International Phonetic Alphabet (IPA)

The International Phonetic Alphabet (IPA), is an alphabet developed with the intention of enabling students and linguists to learn and record the pronunciation of languages accurately, thereby avoiding the confusion of inconsistent, conventional spellings and a multitude of individual transcription systems. One aim of the IPA was to provide a unique symbol for each distinctive sound in a language—that is, every sound, or phoneme, that serves to distinguish one word from another.

IPA primarily uses Roman characters; other letters are borrowed from different scripts (e.g., Greek) and are modified to conform to Roman style. Diacritics are used for fine distinctions in sounds and to show nasalization of vowels, length, stress, and tones. The concept of IPA was first broached by Otto Jespersen in a letter to Paul Passy of the International Phonetic Association and was developed by A.J. Ellis, Henry Sweet, Daniel Jones, and Passy in the late 19th century.

1. The IPA can be used for broad and narrow transcription. For example, in English, there is only one t sound distinguished by native speakers. Therefore, only one symbol is needed in a broad transcription to indicate every t sound. If there is a need to transcribe narrowly in English, diacritic



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marks can be added to indicate that the t's in “tap,” “pat,” and “stem” differ slightly in pronunciation.

2. **STOPS.** - [p]-Pill, [t]-Tilt, [k]-Kilt, [m]-Mild, [n] -Night.
3. **FRICATIVES.** - [f]-Faint, [v]-Vet, [s]-Seal, [z] -Zeal. (Among others, ie. : *The bilabial fricatives, Retroflex fricatives, Velar fricatives, Palatal fricatives, Uvula fricatives, Pharyngeal fricatives & Glottal fricatives*).
4. **AFFRICATES.** - Examples in the words : **Each** & **Judge**.
5. **LATERALS.** - Example: In the production of [l] in the word **lead**, the tongue feels colder on the sides of the tongue that aren't in contact with the roof of the mouth. This is an indication, that air is escaping on the sides.
6. **TRILLS & TAPS/FLAPS.** - Example of a 'Tap': **Butter**, this is as a result of a rapid flick of the tongue at the alveolar ridge. Example of a 'Trill': In the Spanish word '*Perro*', which is an alveolar trill, whereby the tongue is hit many times against the alveolar ridge, this vibration is caused by a flow of air.
7. **APPROXIMANTS.** - Example :[j] in the word '**You**', the tongue and the hard palate are close to each other but they form an open stricture which makes it possible for air to escape without causing an audible friction sound. Other approximants are: [w] - *Velar approximant*, [U] -*Labio - dental approximant, Retroflex approximant*.

THE PRODUCTION OF VOWELS

6.1 Introduction

As we noted before, sounds are classified into consonants and vowels. We now focus on vowels.

It is important to note that

All vowels are produced with a stricture of open approximation.

All vowels are produced with closed glottis. Therefore, vowels are inherently voiced.



Place and manner of articulation are not used to characterize vowels. This is because there is no noticeable obstruction of the airstream during their production. It is not easy to ‘feel’ exactly where they are articulated (Katamba, 1989).

6.2 Lesson Learning Outcomes

By the end of this lesson, you will be able to:

6.2.1 Identify ways of describing vowels in speech

6.2.2 Describe individual vowels

6.2.3 Draw a vowel trapezium

6.2.4 Identify and describe diphthongs

6.2.5 Identify and describe triphthongs

Vowels are described on the basis of the configuration of the tongue in the oral cavity. The shape of the lips is also important in the description.

During the production of most vowels, the tongue forms some sort of a ‘hump’. In regard to the configuration of the tongue in the oral cavity, the tongue could be raised or lowered, or be to the front or to the back of the oral cavity. Two parameters for describing vowels are noted in this case: vertical position and horizontal position

1. Vertical Position of the Tongue Hump

This refers to the relative height of the tongue hump in the oral cavity. There are three positions: high, mid and low.

High vowels are produced with the hump at the highest point.

Examples: [i], [I], [u], []ʊ

Low vowels are produced with the hump at the lowest position.

Examples: [æ], [], [aɛ]

Mid vowels are produced when the hump is in the intermediate position.

Examples: [e], [], []ɔ ε

2. Horizontal axis of the Tongue Hump

The tongue hump may be at the front, center or at the back of the oral cavity.

Examples:



Front Vowels: [i], [I], [e], [æ]

Back Vowels: [u], [], [o], [a]ʊ

Central Vowels: [ə], [], []ɜ e

3. Shape of the Lips

This is the third criterion for describing vowels is lip shape; there are two shapes that the lips assume:

i) Round: some vowels are produced with the lips rounded. Examples: [u], [o], [], ɑ [] ɒ ii) Non-round also referred to as spread are produced with the tongue spread: examples of spread vowels are [i], [I], [e], [æ]

6.2.2 Diphthongs

These are vowels whose quality changes during their production. Let us consider some of the definitions of a diphthongs as given by a number of phoneticians:

- Clark, Yallop and Fletcher (2007) define a diphthong as a segment characterized by articulatory movement of the tongue that is defined in terms of two vocalic targets that starts at one point and end at another.
- Gimson (1980) defines diphthongs as vowels that have a considerable and voluntary glide during their articulation.
- Abercrombie (1967) defines a diphthong as a vowel of continually changing quality while Roach (2004) defines diphthongs as sounds that consist of a glide from one sound to another. Clark, Yallop and Fletcher (1990) on the other hand points out that ‘the glide component is so prominent even though it is still heard as a single sound’

Diphthongs are divided into two categories: centering and closing. Closing diphthong are characterised by a glide towards the high (close) vowels or the edge of the vowel trapezium

Sound	Word	IPA
/ei/	Lay	/lei/
/ou/	Bout	/ bout /
/ai/	Fly	/ flai/
/au/	Bough	/ bau/
/ɔɪ/	Toy	/tɔɪ/

Centring diphthongs are characterised by a glide towards the centre (central vowels) of the vowel trapezium.

Sound	Word	IPA
/iə/	Beer	/biə/
/eə/	Pair	/peə/
/ʊə/	Poor	/puə/

6.2.3 Triphthongs

A triphthong is a glide from one vowel to another and then to a third, all produced rapidly and without interruption. For example, a careful pronunciation of the word 'hour' **auə** begins with a vowel quality similar to [a:], goes on to a glide towards the back close rounded area (for which we use the symbol u :), then ends with a mid-central vowel (schwa, ə). We use [ə] the symbols to represent the way we pronounce 'hour'. The triphthongs can be looked on as being composed of the five closing diphthongs described in the last section, with a added on the end. Thus we get:



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/aɪə/

This triphthong starts at the open region, goes to the front close region and ends at the middle of the mouth. The tongue too has a complex movement during the articulation as it moves from the low position to the high position and finally to the neutral position. The triphthong may appear as:

ire: fire, hire, tyre, buyer, tyre ie: pliers, crier, society
igher: higher, buyer ai: trial, liar iro, ion: iron, lion,
pavilion

2. /aʊə/

This triphthong also starts at the open region, moves to the back close region and ends at the middle of the mouth. The triphthong appears as: our:
our, hour, flour ower: power, shower, flower

3. /e əɪ/

The triphthong begins to move from the front half-close region rising slightly to the front close region then ends at the centre of the mouth. It is another complex triphthong as it involves a multiple configuration of the jaws and the lips. It appears as:

a/e -yer: player, greyer

4. /ɔɪə/

This triphthong moves from the back close position to the front half-close region and ends at the middle of the oral cavity. It has a complex configuration, too, and appears in the following orthographical forms:
oyer: employer

7.2.1 Secondary articulation

Trask (2005) points out that there are two perspectives of defining secondary articulation. The first is the narrow perspective that considers a secondary articulation as any articulatory gesture that complements the primary

articulatory gesture which involves a less constriction than the primary articulatory gesture e.g. labialization or velarization.

The second perspective is the broad perspective, which considers secondary articulation as any of various independent articulatory gestures that can be performed simultaneously with another articulation such as nasalization, prenasalisation, post nasalisation, laryngealisation, pharyngealisation, palatalization, glottalization.

7.2.2 Nasalisation

a) Nasalisation of vowels

Matthew (2005:237) points out that nasal sounds are produced with the lung of the soft palate enabling air to pass through the nose. According to Crystal (1988) the term “nasal” is used in “phonetic classification of speech sounds on the basis of manner of articulation. When a vowel is nasalized, the tilde /~/ or nasalization mark is placed over it.

According to Roach (2017) a nasalized vowel is one that is produced when air escapes through the nose which is quite different from a nasal vowel. French is an example of a language that has nasalised vowel phonemes. A majority of languages, however, have allophonic nasalisation when a vowel occurs close to a nasal consonant. In English, for example, the $\alpha:$ vowel in ‘can’t’ $k\alpha:nt$ is nasalised so that the pronunciation is often (phonetically) $k\tilde{\alpha}:t$.

Prenasalisation

Prenasalisation occurs when the first part of a segment is a nasal followed by a segment of a different value which is not a nasal. Mwita (2007) argues that pre-nasalised stops involve a change from velic opening to velic closure in the production of the two segments in what appears as the same segment length.

Source: Chacha (2007)



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IPA symbol

The IPA uses a ligature that links the two segments together e.g. [mb̥] [d̥] n̥ [ŋk̥] to represent prenasalised segments on the IPA chart (2005).

Examples

Dholuo has five pre-nasalised segments that are shown below:

	IPA	Dholuo orthography	word
Bilabial Prenasalised stop	mb̥	mb	Mbasna 'agemate'
Alveolar Prenasalised stop	nd̥	nd	Ndako 'click'
Palatal prenasalised	n̥ʃ	nj	Njofni 'tapeworm'
Velar Prenasalised	[ŋ̥g]	ng	Ngege
Affricate Prenasalised	[nd̥d͡ʒ]	ndh	'tilapia' Ndhiayo 'far away'



2. Labialisation

According to (Maddieson 1984, Ladefoged and Maddieson 1996, Stonham and Kim 2008) 'labialization' is the additional lip rounding gesture to a segment with the accompanying elevation of the tongue back. There are two types of labialisation:

- i) Simultaneous labialisation
- ii) Transitional labialisation

i) Simultaneous Labialisation

Simultaneous labialization is when the primary segment is accompanied by additional protrusion or rounding of the lips. The additional lip rounding may bring about a difference in meaning in which case we say it is phonemic when it contrasts with a neutral or spread lip posture.

Simultaneous labialisation can be used contrastively or may be only contextual labialisation. For example, consonants adjacent to rounded vowels may be rounded through coarticulation with the lip posture of that sound. For example, in English [s] will most often be labialised [s^w] when ^w surrounded by rounded vowels (eg.u:) in sequences such as "two suits".

IPA symbol

The IPA uses a superscript "w" symbol to the right of the sound being ^w labialised.

e.g. [s^w], [k^w], [l^w]

ii) Transitional Labialisation

In transitional labialisation the lip rounding or protrusion occurs at the end of the primary segment that is being produced. The phonetic environment that triggers transitional lip rounding is when the primary segment occurs before a lip rounded or protruded sound for example [w] or [u]. The difference between the segments is not phonemic and the unlabialised sounds are allophones of the unlabialised counterpart.



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IPA symbol

The IPA uses a superscript "w" symbol [X^w] to the right of the sound being labialised. eg. [s^w], [k^w], [l^w]

Examples

The following examples are probably of simultaneous labialisation.

According to Ladefoged and Maddieson (1996) *Kwakw'ala* Labialised and non-labialised Velar and Uvular stops. Kwakw'ala a language spoken in Vancouver Island, British Colombia Canada. Voiceless velar and uvular oral stops

['kasa]	"beat soft"	[k ^w e sa ^w ']	"splashing"
[qe sa ^w ']	"coiling"	[q ^w e sa ^w ']	"peeling"

Voiced velar and uvular oral stops

['g gis as]	"incest"	[g ^w 'e su]	"pig"
['G Ga as]	"grandparent"	['G ^w alas]	"lizard"

Voiceless velar and uvular fricatives

[xe sa ^w ']	"lost"	[' ^w x asa]	"a dance"
['χasa]	"rotten"	[' ^w 'χ at a]	"sparrow"

Voiceless velar and uvular ejective stops

[' ^w k ata]	"writing"	[k ^w e ^w 'sa]	"light (weight)"
[' ^w q asa]	"sea otter"	[' ^w q asa ^w ']	"crying"

]

7.2.3 Palatalisation, velarisation and Pharyngealisation



Palatalisation is a process where consonants acquire secondary palatal articulation or shift the primary place to, or close to the palatal region. For example, in the word key

/ki:/ [kʲi:]

The English language has three processes that represent palatalisation as shown below:

i) **Coronal palatalisation:** this involve the alternation between alveolar [t,d,s,z] to [, tʃ dʒ z,]

ii) **Velar softening** this involves the alternation between [k, g] and [s, dʒ]

i) **Spirantization:** this involves the alternation between [t]and [s] or [].f

IPA symbol

The IPA standard recommends a superscript "j" following the symbol of the sound being palatalised. This is identical to the transcription of simultaneous palatalisation.

e.g. [l] [p] [k]^j j j

Ladefoged and Maddieson (1996) give the contrast between plain and palatalized labial stops in Russian.

[p o t]	"sweat	[p oʲ tʲ]	"Peter "	[pjo t]	"drin k"
[r aʲ b]	"	[r aʲ bʲ]	"ripple s"		
	"pockmarke	]			
	d"				

Source: Ladefoged and Maddieson (1996)



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4. Velarisation

Crystal (2008) defines velarisation as a secondary articulation involving a movement of the back part of the tongue towards the velum. A velarized sound is a primary sound that has been produced at a different location, usually an alveolar such as [s] that requires the back of the tongue to be raised. The resulting distinctive back sound or 'dark' resonance is clear and the term is applied to consonants other than velar sounds. The velarization may be phonemic in a language such as Arabic that has a distinction between velarized and non-velarized s (transcribed as [p] and [s] respectively). In English, velarization is dependent on context: syllable-final [], as in cool, is given a velar resonance; this can be compared with syllable-initial l, as in leap, where the back of the tongue is much further forward in the mouth (towards the palate). Only simultaneous velarisation is possible for sounds with lingual primary articulations as velarisation involves an adjustment of the entire tongue body. Contrastive velarisations are always simultaneous.

IPA symbols

The IPA standard recommends a superscript " " following the symbol of the sound being velarised.

eg. [s] [t] [d]^ʷ ^ʷ ^ʷ

Also, a tilde character "~" is sometimes placed through another symbol to indicate velarisation or pharyngealisation. This is traditionally done most often to indicate a "dark" or velarised [l] i.e. [l]̤

Examples

According to Ladefoged and Maddieson(1996) in Marshallese a language spoken in the North Pacific has a contrast between normal and velarised bilabial nasal stop.

[mat

"eye"

[m a^ʷ "eel, worm



$$j] \quad t]j \quad "$$

Normal and velarised lateral approximant

[lale	"chec	[ʔ ʔa]	"knoc	[la	"eart
]	k"		k"	ʔ]	h"

5. Pharyngealisation

Pharyngealisation of consonants or vowels occurs when the pharynx or glottis is constricted during the articulation of a particular sound.

The Only simultaneous secondary pharyngealisation appears to be used contrastively in human language. It may be possible, however, for coarticulatory forces to result in transitional pharyngealisation of a bilabial such as [b] when followed (probably across a word or syllable boundary) by the pharyngeal fricative [ʕ] in a language such as Arabic.⁵

IPA symbol

The IPA standard recommends a superscript " " following the symbol of the ɣ sound being pharyngealised. eg. [s] [t] [d]^ɣ ɣ ɣ

Examples

According to Ladefoged and Maddieson (1996) Arabic has a number of normal and pharyngealised ("emphatic") consonant pairs (nb. for some Arabic dialects the secondary articulation is velar).

The consonant pairs are [s, s̥], [z, z̥], [t, t̥], [d, d̥].
Below are examples for [s, s̥] only.

			name
	"to		
[sad]		[sʰ]	of the
	prevail" letter		

Source: Iadefoged and Maddieson (1996)

Even (Siberia) has pharyngealised vowels



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[isli]	"plucked"	[i sli ^{cc}]	"reached"
[us]	"weapons"	[u s ^c]	"guilt"

Phonetics vs Phonology

While phonetics is the study of the inventory of all speech sounds which humans are capable of producing, phonology is the study of the sound systems of languages. Out of the very wide range of sounds the human vocal apparatus can produce (studied by phonetics) only a relatively small number is used distinctively in any one language (phonology).

Ohala (1990) defines phonology as the study of the logical and functional structure and behaviour of the speech sound of a given language X. The phonologist is supposed to answer questions such as: how does language X differentiate between words? why are there cross-language regularities in the sequencing of sound within syllables as well the pattern of assimilation of medial sounds? Phonology is the mental representation of words as well as the knowledge language speakers have of the relationships between the words. Phonetics, on the other hand, studies how sounds are produced, transmitted and perceived by human beings in the process of communication.

There are four main differences between phonetics and phonology as areas of linguistic investigation.

First, both areas seek to answer different questions based on the same input: sounds of the language. The phonetician asks the following questions: What are the sounds? How are they made in the vocal tract? etc. The Phonologist, on the other hand, asks the following questions: How do sounds combine? How do they affect each other? Etc.

The second difference is their object of investigation. Phonetics deals with the physical properties of the elements of the sound system, i.e. how the sound is physically produced. Phonology deals with the sound systems of languages. It asks questions such as: how is speech organized into systems in different languages? How are sounds combined? What is the relation between the different sounds? How do they affect each other? This is a description of the systems and patterns of speech sounds in the language under study.

Third, phonology is concerned with abstract or mental aspects of speech sounds unlike phonetics which will deal with the physical properties. For example, phonetics will describe the properties of the sound [t]- voiceless alveolar stop. Phonology, on the hand, will deal with different instances of the sound in the following English words: ‘tuck’, ‘stuck’, ‘cut’ and ‘duct’. The knowledge speakers possess about the phonological rules in their language is the subject of phonology. For example, which sound sequences might be a word in our language: thrim/blamp vs. gdit/rpukn. How to pronounce words we have never heard before, change foreign words to pattern like the words in our language.

Fourth, the phonologists are interested in the meaning distinctions in the sound system of a language. They want to know:

- i) The function of sounds in the language (what are the phonemes?)
- ii) The way sounds can be combined
- iii) How the sounds are realized in different contexts.

To the phonologist, /r/ can be realized as [r] or as [R] depending on context. However, these are two different sounds to the phonetician, [r] is an alveolar trill and [R] is a uvular trill.

Phones, Phonemes and Allophones:



Phone

A phone is a general term for a unit of speech sound. It is the smallest, perceptible, discrete segment of sound in the stream of speech. Fromkin, Rodman and Hyams (2014) define the phone as the phonetic realization of a phoneme. It may also refer to any speech sound or gesture without regard to its place in the phonology of a language. Phones are the physical realization of phonemes.

Crystal (2008) defines the phone as the smallest perceptible discrete segment of sound in a stream of speech. He adds that, from the viewpoint of segmental phonology, phones are the physical realization of phonemes.

Trask (2005) defines the phone as a single phonetic segment, viewed in terms of its phonetic character and without regard to its possible phonological status.

The collections of phones that are the realizations of the same phoneme are called the allophones of that phoneme. Phonic varieties of a phoneme are referred to as allophones.

Phoneme

There are three ways that one could use to define a phoneme:

- i) **Psychological perspective:** phoneme is the sum total of phonetic properties that all together constitute an indivisible unit. The sound is characterised by duality which is differences between a physiologically concrete sound realization and its mental image. The Phoneme is a psychological sound unit, the sound “intended” by the speaker and “understood” by listeners. Baudouin de Courtenay (1870) defines the phoneme as a mental reality, as the intention of the speaker or the impression of the hearer, or both.
- ii) **Physical perspective:** the phoneme is defined as a class or family of closely related sounds that share certain physical features in common. Daniel Jones (1918): phoneme is a head term of a family of speech sounds.
- iii) **Functional perspective:** the phoneme is defined as a meaning distinguishing unit. According to Trubetzkoy (1926) phoneme is the sum of the phonologically relevant properties of a sound. A phoneme is a minimal unit that can distinguish meanings. The contrast between speech sounds (contrastive distribution); commutation test:

- a) minimal pairs: ran, man, tan, can, ban, fan
- b) near minimal pairs: mission vs. vision

There are other definitions of the phoneme as follows:

Crystal (2008) defines a phoneme as follows:

the minimal unit in the sound system of a language, according to traditional phonological theories. The original motivation for the concept stemmed from the concern to establish patterns of organization within the indefinitely large range of sounds heard in languages. The phonetic specifications of the sounds (or phones) heard in speech, it was realized, contain far more detail than is needed to identify the way languages make contrasts in meaning.

Trask (2005) points out that in many theories of phonology, the phoneme is often a fundamental unit of phonological structure, an abstract segment which is one of a set of such segments in the phonological system of a particular language or speech variety. Trask (2005) defines the phoneme as ‘the smallest unit which can make a difference in meaning’. There are two factors that stand out in the definition of the phoneme:

- i) A phoneme is a unit of contrast where a substitution of one phoneme with another brings a difference in meaning and hence the two phones as assigned as different phonemes.
- ii) Phonemes are identified and assigned in reference to a particular speech variety. Languages differ in the number of phonemes they distinguish.

Jones (1967: 258) considers the phoneme to be a “family” of sounds in a particular language that “count for practical purposes as if they were one and the same.”

Bloomfield (1933: 77–78) characterized the phoneme as:

“a minimum unit of distinctive sound-feature . . . The speaker has been trained to make sound-producing movements in such a way that the phoneme features will be present in the sound waves, and he has been trained to respond only to these features [\[JM1\]](#) [\[JM2\]](#) .”

9.2.2 Discovery Procedures for Phonemes

According to Trubetzkoy (1939) Phonemes are the linguistically contrastive or significant sounds, or set of sounds, of a language. This contrast which



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distinguishes meaning is known as a phonemic distinction. Linguistic **units** which cannot be substituted for each other without a change in meaning can be referred to as **linguistically contrastive** or significant units. Such units may be phonological, morphological, syntactic, semantic etc. Logically, this takes the form:-

IF	unit X	in context A	GIVES meaning 1
AND IF	unit Y	in context A	GIVES meaning 2
THEN	unit X AND unit Y		belong to separate linguistic units

eg. IF sound [p] in context [_ n]ɪ GIVES meaning "pin"
 AND IF sound [b] in context [_ n]ɪ GIVES meaning "bin"

belong to separate

THEN sound [p] and sound [b]

linguistic units

Minimal pair test

McMahon (2002) defines a minimal pair as the substitution of one sound in a word with another to see if there is a difference in meaning. The diagnostic test he give is the substitution of the sound in a word in the context [_at] in English that gives the phonemes [p, b, m, f, v, ð, t, s, n, r, , k,tʃ h].

Minimal pairs are words whose sound structures are identical except one minimal difference, a single sound segment that occurs in the same place in the string.

Substituting one of the sounds for another creates a different word with a totally different meaning, substituting a different vowel sound in the following English words:

	crick	/kr k/ɪ	
creek	/kri:k/		crook
/kr k/ʊ		croak	
/krə k/ʊ		crack	



/kræk/

crock /kr k/ɒ

In Kiswahili the words *papa*, [papa] ‘type of fish’ *baba* [baba] ‘father’ and *mama* [mama] ‘mother’ are a minimal pair since the difference in the bilabial sound causes the difference in meaning between the words in question. Minimal pairs exist in a number of languages and they exhibit both vowel and consonant contrast in language in the English language for example the following word would illustrate:
 cap /kæp/ cup /kʌp/

A minimal pair can also be a sentence that has only a single sound as being different and that will cause a difference in the meaning of the two sentences. For example:

Which hut/hat is yours? is a minimal pair illustrating the /ʌ/ vs /æ/ contrast in English.

In Dholuo the following set of words illustrate minimal pairs:

Word	Transcription	gloss
Pi	/pi/	‘water’
Bi	/bi/	‘come’ imperative
Mbi	/mbi/	‘charisma’
Ki	/ki/	‘wait’
Ji	/dʒi/	‘people’
Ngi	/ŋi/	‘look’ imperative
Ti	/ti/	‘work’ imperative
Dhi	/ði/	‘go’ imperative

Near minimal pair test / contrast in analogous environment

Mannell (2009) points out that Contrast in Analogous Environment (CAE) is a near minimal pair where more than one sound differs. The differences between the



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words cannot be explained by the difference caused by the adjacent sounds. The other points of variation in the pair of words are as remote as possible (and certainly never adjacent and preferably not in the same syllable) from the environment of the pairs of sounds being tested. Hayes (2009) gives the following example of near minimal pairs in English.

Hayes (2009:35)

Allophones

Trask (2005) defines the allophone as one of two or more phonetically distinct segments which can realize a single phoneme in varying circumstances.

Allophones are the linguistically non-significant variants of each phoneme. This is a phoneme that may be realized by more than one speech sound depending on the position of the word where the sound occurs. Within a given language, some sounds are considered to be the same sound, even though they are phonetically distinct.

Allophones of /p/

- decide whether the two sounds of /p/ are the same or different in the words below?
- pool [p^hul] vs. spool [spul]
 - phonetically different (aspirated vs. unaspirated)
 - native speakers perceive the same sound

Allophones of /t/

A phoneme can be pronounced in different ways according to its context.

Compare:

- a) The difference between /t/ in : tea, eat, writer, eighth, two
- b) The difference between /i:/ in: see, seed, seat, seen

- Therefore, a phoneme may have more than one realization.
- The different realizations of a phoneme are called allophones of that phoneme. The allophone is a variant of a phoneme.



[p ^h u:]†	pool
[spu:]†	spool
[k ^h I]†	kill
[skI†]	skill

The aspirated sounds are in complementary distribution with the plain counterpart.

Two or more sounds are allophones of the same phoneme if:

- a) They have a predictable, complementary distribution;
- b) They do not create a semantic contrast; and
- c) They are phonetically similar.

(E.g. [l] and [ɫ] in English: [l] never occurs before consonants or word-finally, [ɫ] never occurs before vowels) The case of “t” in American English

Say the following words:

top, stop, metal, and right

What is the difference?

top [t^hap] the “t” is aspirated [t^h]

stop [stap] the “t” is unaspirated [t]

metal [mɛrl] the “t” is a flap [ɾ]

right [raɪt'] the “t” is unreleased [t']

This different position where a sound may occur is known as the phonetic environment of the phoneme which may depend on the variety of language that one speaks which is known as the dialect. The choice of an allophone may be conditioned by the idiolect of the individual's unique way of pronunciation which



varies from selection is person to person and occasion to occasion i.e. free variation.

What is prosodic phonology?

Crystal (2008) points out that the term suprasegmental is used in phonetics and phonology to refer to a vocal effect which extends over more than one sound segment in an utterance, such as a pitch, stress or juncture pattern. In its contrast with ‘segmental’, it is seen as one of two main classes into which phonological units can be divided.

Trask (2005) gives three dimensions in his definition of prosody as follows:

- i) The study of stress, pitch and intonation.
- ii) A phonological element which is realized phonetically on more than one segment, including for example (in certain circumstances in certain languages) liprounding, backness or nasalization. The extended realization of height, backness, lip-rounding or ATR in vowel harmony languages in Nilotic languages is an outstanding example.
- iii) A phonological element which can only be described with reference to a domain longer than a single segment such as stress and tone.

Trask (2005) adds that prosodic elements can also be referred to as suprasegmental element whose domain is something larger than a single segment and whose phonetic realization can only be described by reference to adjoining domains in the same utterance. The most familiar such elements are stress and tone (or pitch), though others are sometimes recognized.

10.2.2 Syllable

Roach (2000:70) define the syllables phonetically as follows “ as consisting of a centre which has little or no obstruction to airflow and which sounds

comparatively loud; before and after that centre (...) there will be greater obstruction to airflow and/or less loud sound” (Roach, 2000: 70). In the monosyllable (one-syllable word) *cat* /kæt/, the vowel /æ/ is the “centre” at which little obstruction takes place, whereas we have complete obstruction to the airflow for the surrounding plosives /k/ and /t/.

Laver (1994: 114), on the other hand, defines the syllable phonologically as “a complex unit made up of nuclear and marginal elements”. Nuclear elements are the vowels or syllabic segments; marginal elements are the consonants or nonsyllabic segments. In the syllable *paint* /pe nt/, the diphthong /e / is the nuclear element, while initial consonant /p/ and the final cluster /nt/ are marginal elements.

There are three theories that explain the syllable:

- i) According to the prominence theory, for example, which is based on auditory judgements, the number of syllables in a word is determined by the number of peaks of prominence. In the word *entertaining* / entə te n ɪ / the peaks of , ' ɪ ɪ prominence are represented by the vowels /e ə e /.ɪ ɪ However, this theory does not help much in discussions of syllable division ii) The chest pulse theory discusses the syllable in the context of muscular activities and lung movements in the process of speech. Experiments have shown that the number of chest pulses, accompanied by increase of air pressure can determine the number of syllables produced (Gimson, 1980: 56), thus allowing to associate the number of syllables with the number of chest pulses. This approach, however, cannot account for cases when 2 vowels occur one after the other – for example in words like *being* / bi: ɪ / or ' ɪ *playing* / ple ɪ / the second chest pulse ' ɪ might be almost irrelevant and thus lead erroneously to the conclusion that such English words consist of one syllable only.
- iii) Sonority theory according to which the pulses of pulmonic airstream in speech “correspond to peaks in sonority” (Giegerich, 1992: 132). The sonority of a speech sound is discussed as “its relative loudness compared to other sounds” (Giegerich, 1992: 132) and each syllable corresponds to a peak in the flow rate of pulmonic air. Thus nuclear elements, or syllabic segments can be described as intrinsically more sonorous than marginal, or non-syllabic elements.

Speech sounds can be ranked in terms of their intrinsic sonority according to a sonority scale. The sonority scale for English is given below (although in



principle it is also valid for other languages). Voiced segments are more sonorous than voiceless ones and sonorants are more sonorous than obstruents; vowels are more sonorous than consonants, open vowels being more sonorous than close ones. The disyllabic word *painting* / pe nt ɪŋ/ has been plotted onto the sonority scale as an example.

The syllable is one of the basic of the components of organization above the phoneme that is critical in the phonological organization of language.

‘A syllable is a phonological unit composed of one or more phonemes. Every syllable has a nucleus, which is usually a vowel (but can be a syllabic liquid or nasal). The nucleus may be preceded and/or followed by one or more phonemes called the syllable onset and coda.’ Fromkin, Rodman and Hyams (2003)

‘The syllable phonetically is considered as having a center which has little or no obstruction and which sounds comparatively louder than before and after the center. A minimum syllable will be a single vowel in isolation ‘are’ [a:], ‘or’ [:] and ‘err’ [:]. Some syllables have onset ‘bar’ [ba:], ‘key’ [ki:] and ‘more’ [m :].’ 3 Roach (1983)

1. Every syllable is made up of (at most) three parts:

i) Onset: the consonants that begin the syllable ii)

Nucleus: the sound in the middle of the syllable (usually a vowel) iii)

Coda: the consonants the end the syllable

2. Syllables can differ in size:

i) Some syllables do not have onsets (e.g. [it])

ii) Some syllables do not have codas (e.g. [ti:])

iii) But, every syllable has a nucleus (e.g [ai])

The syllable has hierarchical structure as shown below:

Fig. 10.1 Syllable of the English word ‘cat’

The figure above of the English word cat /kæt/ has the following constituents, the Onset, Peak and Coda each consist of one segment: the consonant (C) /k/ occupies the Onset, the vowel (V) /æ/ – the Peak, and the consonant /t/ is the Coda of this

syllable. Fromkin, Rodman and Hyams(2014) show the maximal syllable structure as illustrated below:

The syllable structure of the word *splints* is shown below

Rime (American English)

Fromkin, Rodman and Hyams (2014) summarises the major syllable structures in English below

10.2.3 Stress

Ladefoged and Johnson (2011) define a stressed syllable as one that has greater respiratory energy than neighbouring unstressed syllables. It may also have an increase in laryngeal activity. Stress can always be defined in terms of something a speaker does in one part of an utterance relative to another.

According to Katamba (1989), stressed syllables tend to have higher pitch and longer duration than their unstressed counterparts. The main phonetic ingredients of stress are pitch, length and loudness. Phonologically, there are several kinds of stress, one of which is word stress. From the perspective of the listeners a syllable is stressed, when it is pronounced with the following: i) longer in duration ii) higher in pitch iii) louder in volume

In every language, one or more syllables in content words are stressed and this is indicated by ()` as shown below.

You will notice that the location of the primary word stress in these words determines their word class and meaning. Some words may contain more than one stressed vowel, but exactly one of the stressed vowels is more prominent than the others. The vowel that receives primary stress is marked by an acute accent. The other stressed vowels are indicated by a grave accent (`) over the vowels (these vowels receive secondary stress).

Source Fromkin, Rodman and Hyams

(2011)



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According to Fromkin, Rodman and Hyams (2011) stress is a property of the syllable rather than a segment; it is a prosodic or suprasegmental feature. To produce a stressed syllable, one may change the pitch (usually by raising it), make the syllable louder, or make it longer. We often use all three of these phonetic means to stress a syllable.

Fromkin, Rodman and Hyams (2011) conclude that dialectal variation may account for the difference in stress marks. For example, in most varieties of American English the word *laboratory* [lábərət^həri] has two stressed syllables, but in most varieties of British English it receives only one stress [ləb rətri].^ɔ

10.2.4 Length

The sounds of a language may be said to be either long or short. In English, the contrast between [i:] and [i] as in ‘sheep’ and ‘ship’ can be seen in terms of vowel length. In this case, [i:] is said to be long while [i] is a short vowel. To represent vowel length, two dots, [:] are placed after the vowel. In some languages, for example Luanda, consonants vary in length. An example is the word ‘tta’ [t:a] ‘kill’ contrasted to ‘ta’ [ta] ‘release’.

According to Vidali and Mubanga (2014), Bemba, a Bantu language spoken in Zambia, has a contrastive semantic distinction between short and long vowels (the doubling of vowels represents vowel length):

Source Vidali and Mubanga (2014)

Long vowels may also result from the fusion of two vowels across morpheme boundaries. For example, umwééní ‘stranger, guest’ is derived from umu– + –eni (class 1 singular + stem), and its plural abééní ‘strangers, guests’ from aba– + –eni (class 2 plural prefix + stem).

Ladefoged and Johnson (2011) give the following examples from Italian and Luganda to show that there is contrastive segment length in these two languages. Many other languages make considerable use of length contrasts. Long consonants that can be analysed as double consonants are called **geminate**s.

Language	long consonant	short consonant
Luganda	[k ^h ulà] ‘treasure’	[k ^h ulà] ‘grow up’
Italian	[n nno] grandfather ^ɔ	[n no] ‘ninth’



Italian [pappa] ‘baby food’

[papa] ‘pope’

Source: Ladefoged and Johnson (2011)
